

D419

Implementing and Administering Networking Solutions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A clinic's IT lead is carving subnets from 172.16.0.0/16, which the clinic has used since it opened. A new imaging suite needs 64 usable host addresses, and the vendor quoted a two-day install. What is the smallest subnet mask that fits?

- A. 255.255.255.128 (/25)
- B. 255.255.255.192 (/26)
- C. 255.255.255.0 (/24)
- D. 255.255.0.0 (/16)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

How many usable host addresses does a /27 subnet provide?

- A. 32
- B. 30
- C. 62
- D. 27

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Two switches connect over one link. Hosts in VLAN 10 and VLAN 20 sit on both switches. The link is a new Cat6a run through a freshly renovated ceiling. VLAN 10 hosts can reach each other across the link, but VLAN 20 hosts cannot. The inter-switch port is set as an access port in VLAN 10. What is the fix?

- A. Change the port to access VLAN 20.
- B. Replace the new cable run.
- C. Add a router, because a VLAN can never span two switches without routing.
- D. Configure both ends as 802.1Q trunks that allow VLANs 10 and 20.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Three switches are cabled in a triangle. Switch A: priority 32768, MAC 0000.1111.aaaa. Switch B: priority 4096, MAC 0000.2222.bbbb. Switch C: priority 32768, MAC 0000.0000.0001. Switch A is the newest and fastest model. Which switch becomes the root bridge?

- A. Whichever switch has the highest bridge ID.
- B. Switch A, because it is the newest and fastest.
- C. Switch B
- D. Switch C, because it has the lowest MAC address.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

An engineer wants Switch D, a distribution switch, to become the root bridge. What change makes that happen?

- A. Raise Switch D's priority to 61440.
- B. Replace Switch D with a unit that has a higher MAC address.
- C. Give Switch D the fastest uplinks, since the fastest switch always wins.
- D. Set Switch D's bridge priority lower than every other switch's (for example, 4096).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A router holds three routes: 10.0.0.0/8 via OSPF, 10.1.0.0/16 via a static route, and 10.1.5.0/24 via RIP. The static route was added by a senior engineer last year. A packet arrives for 10.1.5.9. Which route forwards it?

- A. 10.0.0.0/8 via OSPF
- B. 10.1.0.0/16 via static
- C. 10.1.5.0/24 via RIP
- D. All three, load-balanced.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A router learns 192.168.50.0/24 from OSPF with a metric of 20 and from EIGRP with a metric of 3,072,000. The OSPF neighbor sits in the main data center with redundant power. Which route goes into the routing table?

- A. Both, load-balanced equally.
- B. The OSPF route, because 20 is a lower metric.
- C. The EIGRP route.
- D. The OSPF route, because its neighbor has redundant power.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A static route should be used only if the OSPF route to the same prefix disappears. What administrative distance should the static route get?

- A. 100
- B. A value above 110, such as 115.
- C. 0
- D. 1, the default for static routes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An inbound ACL on a branch router reads, in order: (1) permit tcp any host 10.9.9.10 eq 443; (2) deny ip 10.1.1.0 0.0.0.255 any; (3) permit ip any any. The branch has 40 staff and a new fiber link. Host 10.1.1.25 sends HTTPS to 10.9.9.10. What happens?

- A. Denied by line 2, because the source subnet is blocked.
- B. Permitted only because of line 3.
- C. Denied by the implicit deny.
- D. Permitted by line 1.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

An inbound ACL contains a single line: deny ip host 10.1.1.5 any. What happens to traffic from 10.1.1.6?

- A.** It is denied by the implicit deny at the end of the list.
- B.** It is logged and forwarded.
- C.** It is permitted, because ACLs default to permit.
- D.** It is permitted, because it does not match the deny line.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A warehouse deploys three 2.4 GHz access points with overlapping coverage. The APs are ceiling-mounted at 30 feet and powered by PoE. Which channel plan avoids overlap?

- A.** Channels 1, 6, and 11.
- B.** Channels 1, 2, and 3.
- C.** All three on channel 6 so clients roam easily.
- D.** Any three, since 2.4 GHz channels never overlap.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A small office has 60 devices on 192.168.1.0/24 and one public IPv4 address from its ISP. The office manager wants the work finished before a Monday audit. Which setup lets all devices reach the internet at the same time?

- A.** No NAT; private addresses route on the internet.
- B.** PAT (NAT overload), sharing the public address by port number.
- C.** Dynamic NAT with a pool of that one address.
- D.** Static NAT for each device.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A clinic's IT lead is carving subnets from 172.16.0.0/16, ~~which the clinic has used since it opened~~. A new imaging suite needs 64 usable host addresses, ~~and the vendor quoted a two-day install~~. What is the smallest subnet mask that fits?

Target word: smallest mask for 64 hosts **Decides:** A /26 gives only 62 usable, so 7 host bits (/25) are needed.

A	Correct	$2^7 - 2 = 126$ usable, the smallest that holds 64.
B	Dropped element	$2^6 = 64$ addresses, but minus network and broadcast leaves 62.
C	Adjacent category	Fits, but 254 usable is not the smallest option.
D	Camouflage	The parent block, not the subnet the suite needs.

2. Answer B Concept 1 · transfer

How many usable host addresses does a /27 subnet provide?

Target word: solve for hosts from a prefix **Decides:** $32 - 27 = 5$ host bits.

A	Dropped element	Total addresses; leaves in the network and broadcast addresses.
B	Correct	$2^5 - 2 = 30$.
C	Adjacent category	That is a /26.
D	Wrong stem number	The prefix length, not a host count.

3. Answer D Concept 2

Two switches connect over one link. Hosts in VLAN 10 and VLAN 20 sit on both switches. ~~The link is a new Cat6a run through a freshly renovated ceiling~~. VLAN 10 hosts can reach each other across the link, but VLAN 20 hosts cannot. The inter-switch port is set as an access port in VLAN 10. What is the fix?

Target word: why one VLAN fails across switches **Decides:** An access port carries only one VLAN.

A	Dropped element	Fixes VLAN 20 but breaks VLAN 10.
B	Camouflage	VLAN 10 works over the same cable, so the cable is fine.
C	Absolute language	Routing is for traffic between VLANs; one VLAN spans switches over a trunk.
D	Correct	A trunk carries multiple tagged VLANs over one link.

4. Answer C Concept 3

Three switches are cabled in a triangle. Switch A: priority 32768, MAC 0000.1111.aaaa. Switch B: priority 4096, MAC 0000.2222.bbbb. Switch C: priority 32768, MAC 0000.0000.0001. ~~Switch A is the newest and fastest model~~. Which switch becomes the root bridge?

Target word: root bridge **Decides:** Priority is compared before MAC; B has the lowest priority.

A	Inverted operation	The lowest bridge ID wins.
B	Camouflage	Speed and age play no part in the election.
C	Correct	4096 is the lowest priority, so B has the lowest bridge ID.
D	Dropped element	MAC only breaks ties; C loses on priority first.

5. Answer D Concept 3 • transfer

An engineer wants Switch D, a distribution switch, to become the root bridge. What change makes that happen?

Target word: solve for the setting that wins root **Decides:** Lowest priority wins.

A	Inverted operation	A higher number makes D less likely to win.
B	Confusion set	A higher MAC loses ties.
C	Absolute language	Link speed affects path cost, not the root election.
D	Correct	A lower priority gives D the lowest bridge ID.

6. Answer C Concept 4

A router holds three routes: 10.0.0.0/8 via OSPF, 10.1.0.0/16 via a static route, and 10.1.5.0/24 via RIP. ~~The static route was added by a senior engineer last year.~~ A packet arrives for 10.1.5.9. Which route forwards it?

Target word: route selection among overlapping prefixes **Decides:** The /24 is the most specific match.

A	Inverted operation	Picks the least specific route.
B	Adjacent category	Applies administrative distance to routes with different prefix lengths.
C	Correct	Longest prefix match is applied first; AD only compares identical prefixes.
D	Confusion set	Load balancing needs equal routes to the same prefix.

7. Answer C Concept 5

A router learns 192.168.50.0/24 from OSPF with a metric of 20 and from EIGRP with a metric of 3,072,000. ~~The OSPF neighbor sits in the main data center with redundant power.~~ Which route goes into the routing table?

Target word: same prefix, different protocols **Decides:** EIGRP AD 90 beats OSPF AD 110.

A	Adjacent category	Equal-cost load balancing happens within one protocol, not across two.
B	Confusion set	Metrics from different protocols are not compared.
C	Correct	Lower administrative distance wins when the prefix is identical.
D	Camouflage	Neighbor hardware does not enter route selection.

8. Answer B Concept 5 • transfer

A static route should be used only if the OSPF route to the same prefix disappears. What administrative distance should the static route get?

Target word: solve for AD of a floating static **Decides:** It must be worse (higher) than OSPF's 110.

A	Dropped element	Raised, but still below 110, so it would still win.
B	Correct	A floating static stays out of the table until the OSPF route is gone.
C	Confusion set	0 is for directly connected networks.
D	Inverted operation	AD 1 would beat OSPF and take over at once.

9. Answer D Concept 6

An inbound ACL on a branch router reads, in order: (1) permit tcp any host 10.9.9.10 eq 443; (2) deny ip 10.1.1.0 0.0.0.255 any; (3) permit ip any any. ~~The branch has 40 staff and a new fiber link.~~ Host 10.1.1.25 sends HTTPS to 10.9.9.10. What happens?

Target word: fate of one packet **Decides:** Line 1 matches first.

A	Dropped element	Ignores that line 1 already matched.
B	Adjacent category	Line 3 is never reached for this packet.
C	Confusion set	The implicit deny only catches traffic no line matched.
D	Correct	The first matching line decides; processing stops there.

10. Answer A Concept 6 · transfer

An inbound ACL contains a single line: deny ip host 10.1.1.5 any. What happens to traffic from 10.1.1.6?

Target word: unmatched traffic **Decides:** Every ACL ends with an implicit deny.

A	Correct	No permit matches it, so the hidden deny any drops it.
B	Confusion set	Logging is optional and does not permit traffic.
C	Inverted operation	The default at the end of an ACL is deny.
D	Dropped element	Forgets the implicit deny.

11. Answer A Concept 7

A warehouse deploys three 2.4 GHz access points with overlapping coverage. The APs ~~are ceiling-mounted at 30 feet and powered by PoE.~~ Which channel plan avoids overlap?

Target word: non-overlapping channels **Decides:** Only 1, 6, and 11 do not overlap in 2.4 GHz.

A	Correct	They are spaced far enough apart not to overlap.
B	Adjacent category	Neighboring channels overlap heavily.
C	Confusion set	Same channel creates co-channel contention.
D	Absolute language	Most 2.4 GHz channels overlap.

12. Answer B Concept 8

A small office has 60 devices on 192.168.1.0/24 and one public IPv4 address from its ISP. ~~The office manager wants the work finished before a Monday audit.~~ Which setup lets all devices reach the internet at the same time?

Target word: many-to-one translation **Decides:** One public address for many hosts.

A	Inverted operation	RFC 1918 addresses are not routed on the public internet.
B	Correct	Port tracking lets many inside hosts share one address.
C	Dropped element	Without overload, only one host can use it at a time.
D	Adjacent category	One-to-one mapping needs 60 public addresses.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Subnet sizing	Usable hosts = $2^{(\text{host bits})} - 2$; pick the smallest prefix whose usable count meets the need.	Forgetting to subtract the network and broadcast addresses.	“Needs N usable host addresses” or a prefix length and “how many hosts.”
2	Access ports vs. trunks	An access port carries one VLAN untagged; a trunk carries many VLANs with 802.1Q tags. Same VLAN across switches needs a trunk, not a router.	Adding routing to fix a VLAN that simply is not allowed across the link.	One VLAN works across switches and another does not.
3	Spanning Tree root election	The lowest bridge ID wins root: priority is compared first, MAC address only breaks ties.	Picking the lowest MAC or the newest, fastest switch.	A list of priorities and MACs.
4	Longest prefix match	A router forwards using the most specific matching route, before administrative distance is ever considered.	Letting a lower-AD but less specific route win.	Overlapping routes of different prefix lengths.
5	Administrative distance	For the same prefix from different sources, the lowest AD wins (connected 0, static 1, EIGRP 90, OSPF 110, RIP 120); metrics are compared only within one protocol.	Comparing metrics across protocols.	One prefix learned two ways.
6	ACL processing	ACL lines are read top down, the first match decides, and an implicit deny ends every list.	Assuming a later, more specific line overrides an earlier match, or that unmatched traffic passes.	An ordered rule list and a sample packet.
7	2.4 GHz channel planning	In the 2.4 GHz band only channels 1, 6, and 11 do not overlap.	Using adjacent channels or one channel everywhere.	Several nearby 2.4 GHz access points.
8	NAT and PAT	PAT (NAT overload) lets many inside hosts share one public address by tracking port numbers; static NAT is one-to-one.	Choosing static or pooled NAT when there is only one public address.	Many private hosts, one public IP.

Course scope (public description): Implementing and Administering Networking Solutions expands on basic networking concepts and covers advanced network engineering skills including: Switch and router configuration, trouble shooting and maintenance on wired and wireless networks, Security, network automation and introduces Software Defined Networking. This course prepares students for the Cisco Certified Network Associate (CCNA) certification exam CCNA-200-301.